

IMO(52nd) – 2011

First Day

1. Given any set $A = \{a_1, a_2, a_3, a_4\}$ of four distinct positive integers, we denote the sum $a_1 + a_2 + a_3 + a_4$ by S_A . Let n_A denote the numbers of pairs (i, j) with $1 \leq i < j \leq 4$ for which $a_i + a_j$ divides S_A . Find all sets A of four distinct positive integers which achieve the largest possible value of n_A .
2. Let S be a finite set of at least two points in the plane. Assume that no three points of S are collinear. A windmill is a process that starts with a line ℓ going through a single point $P \in S$. The line rotates clockwise about the pivot P until the first time that the line meets some other point belonging to S . This point, Q , takes over as the new pivot, and the line now rotates clockwise about Q , until it next meets a point of S . This process continues indefinitely. Show that we can choose a point P in S and a line ℓ going through P such that the resulting windmill uses each point of S as a pivot infinitely many times.
3. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a real valued function defined on the set of real numbers that satisfies $f(x+y) \leq yf(x) + f(f(x))$ for all real numbers x, y . Prove that $f(x) = 0$ for all $x \leq 0$.

Second Day

4. Let $n > 0$ be an integer. We are given a balance and n weights of weight $2^0, 2^1, \dots, 2^n$. We are to place each of n weights on the balance one after another, in such away that the right pan is never heavier than the left pan. At each step we choose one of the weights that has not yet been placed on the balance, and place it either it on the left pan or the right pan, until all of the weights have been placed. Determine the number of ways in which this can be done.
5. Let f be a function from the set of integers to the set of positive integers. Suppose that, for any two integers m and n , the difference $f(m) - f(n)$ is divisible by $f(m - n)$. Prove that for all integers m and n with $f(m) \leq f(n)$, the number $f(n)$ is divisible by $f(m)$.
6. Let ABC be an acute angled triangle with circumcircle Γ . Let ℓ be a tangent to Γ , and let ℓ_a, ℓ_b, ℓ_c be the lines obtained by reflecting ℓ in the lines BC , CA and AB respectively. Show that the circumcircle of the triangle determined by the lines ℓ_a, ℓ_b, ℓ_c is tangent to the circle Γ .

